Exchange of gases by the respiratory system involves three different processes such as;

1. **External Respiration:** Intake of O_2 from the air and releasing of CO_2 from the lungs occurs through nostrils.
2. **Internal Respiration:** Taking in of oxygen and giving out CO_2. The circulatory system transports O_2 and CO_2 to and from all parts of the body. Hemoglobin in the red blood cells (RBCs) transport O_2 and CO_2.
3. **Cellular Respiration:** Cells take in O_2 and release CO_2

Activity 5:

Aim: To prove that exhaled air is rich in carbon- di- oxide

Materials required: Two glass jars with lime water and a straw

Procedure: Leave the first jar with lime water undisturbed, blow air in to the second jar with the help of a straw

Observation: Lime water turns milky in few seconds in the second jar. The CO_2 gas alone can change the lime water into milky white.

Conclusion: Carbon-di-oxide is present in the air that we exhale.



Each lung has about 300 million air sacs or alveoli.

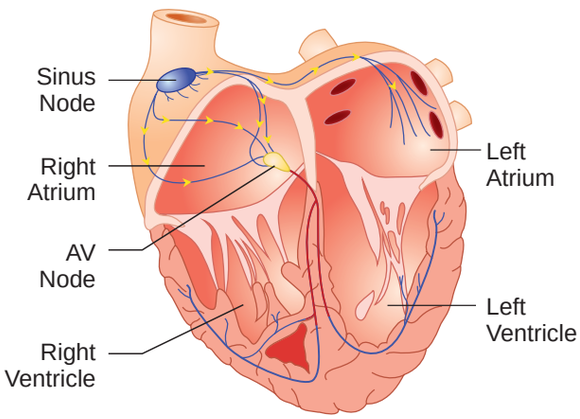
Yawning helps us to take in more amount of O_2 and to give out CO_2.

6.5 Circulatory system

The circulatory system is one of the important system consisting of heart, blood vessels and blood. It transports respiratory gases, nutrients, hormones and waste materials within the body. It protects the body from harmful pathogens and also regulates the body temperature.

Heart

Heart is located in the thoracic cavity between the two lungs. The heart is four chambered and is surrounded by a double layered membrane called **pericardium**. The heart pumps blood continuously throughout our life time.



Blood vessels

Three types of blood vessels are present in the body. They are **arteries, veins and capillaries.** They form a closed network through which the blood is circulated.

Blood

Blood is a fluid connective tissue of red colour containing plasma and blood cells. There are three types of blood cells namely, Red blood corpuscles (RBCs), White Blood corpuscles (WBCs) and Blood Platelets. RBCs are produced in the bone marrow.

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